



AIM-394

VICAT APPARATUS



Aimil Ltd.

Instrumentation & Technologies

Version

1.2



Aimil Ltd.

Automation & Technology

**AIM – 393
VICAT APPARATUS
WITHOUT DASHPOT**

**AIM – 394
VICAT APPARATUS
WITH DASHPOT**

Operating Instructions



Operating Instructions Vicat Apparatus Vicat Apparatus with Dashpot

Introduction

The equipments meet the requirement of IS : 5513. These are used for the determination of Standard Consistency, Initial Setting Time and Final Setting Time of cement paste of Ordinary Portland Cement, Rapid Hardening Portland Cement, Low Heat Portland Cement and other Hydraulic Cements as per the procedure described in IS-1727, 2541 (P.1), 2645, 4031, BS : 12, 146, 915, 1370, 4027, 4246, 4248, ASTM C-91, C-141, C-187, C-191, C-308, C-472 and ASSHO T-129 E-131.

The AIM-394 Model has an additional attachment of a dash pot which facilitates the lowering of a movable rod slowly.

Both these models of the Vicat Apparatus bear the ISI Certification Mark IS : 5513, which ensures and fully guarantees the quality and accuracy of the apparatus.

The narration given under various captions in this text applies to both these models of Vicat Needle apparatus. For AIM-394 in particular, additional remarks for the use of the dash-pot and for the setting up of the needles and the consistency plunger have been furnished.

Description

The equipments are illustrated in the General Assembly Drawing attached. The numbers given against the components in the description below pertain to this General Assembly Drawing.

The instrument consists of a base(1) connected to a metallic frame (6) bearing a movable rod (10) with a cap (11) at one end. At the other end, any one of the following attachments which are interchangeable may be attached.

1. Initial Needle (4)
2. Final Needle (14)
3. Plunger (13)

The movable bearing rod (10) carries an indicator (8) which moves over a graduated scale (7) attached to the frame (6). A release pin (9) is provided for holding the movable rod while fixing the needle or plunger.

In the AIM-394 model, a dash-pot is fitted to the frame. The dash-pot (12) is filled with oil of low viscosity (oil is not supplied) and helps in lowering the movable rod slowly. It is recommended to use 3-in-1 or Singer oil or any equivalent oil of low viscosity.

Vicat Mould

The mould (3) is in the form of frustum of a cone with an internal diameter of 70mm at the lower, 80 mm at the upper end and a height of 40 ± 0.2 mm. The mould is kept on a glass plate (2) which is placed on the base (1) of the Vicat Apparatus.

Initial Needle

The initial needle (4) is circular in cross-section and has a cross sectional area of 1 mm^2 . The end of the needle is flat. This is used for the determination of the initial setting time of the cement sample.

Final Needle

The final needle (14) is circular in cross-section and has a cross sectional area of 1 mm^2 . The needle is fitted in a metal attachment (hollowed out). The end of the needle projects beyond the cutting edge of the hollowed out metal attachment. This is used for determining the final setting time of cement.

Plunger

The Plunger (13) is known as the consistency plunger. It is made of polished brass with a projection at the upper end for insertion into the movable rod. The lower end is flat.

Setting Up

Keep the Vicat Apparatus on a level base. When using the AIM-394 model, keep the movable rod at its highest position and hold it with the pin. Unscrew the top of the dashpot and half fill it with suitable oil of low viscosity and screw on the top back on. Work the plunger a couple of times.



Temperature for Testing

Determination Of Consistency Of Standard Cement Paste

1. Definition

The standard consistency of a cement paste is that which will permit the Vicat Plunger to penetrate to a point, 5 to 7 mm from the bottom of the Vicat Mould, when cement is tested as described below.

2. Specimen Preparation

Take 400g of weighed cement in a dish. Run down a known quantity of water from the burette into the dish and mix it thoroughly taking care that the time of gauging is neither less than 3 minutes nor more than 5 minutes and that gauging is completed before any sign of setting occurs. The gauging time is the time elapsed from the time of adding water to the dry material until the time the user starts to fill the mould. Clean appliances shall be used for gauging. The gauging trowel (AIM-428 Gauging Trowel supplied as accessory at extra cost) and the operator's hand shall alone be used.

3. Test Procedure

Apply petroleum jelly to the inner surface of the Vicat Mould and the Base Plate, so that there is no water leakage. Fit the plunger to the movable rod, prepare the cement paste as described above and fill mould. For filling the mould, the operator's hand and the blade of the gauging trowel shall alone be used. Level-off the surface of the mould, place the whole assembly on the base of the Vicat Apparatus and slowly bring the plunger attached to the bearing rod down to the surface of the cement paste and then quickly release it (When AIM-394 Vicat Needle Apparatus with dashpot is used, place the mould filled with cement paste and the non-absorbent plate on the base plate of the Vicat Apparatus. Raise the plunger of the dashpot, bring it in contact with the top cap of the movable bearing rod, remove the pin holding the movable bearing rod to the surface of the cement paste and quickly release by pushing the plunger of the dashpot down with hand). All these operations are to be carried out within the minimum possible time after filling the mould, so that there may not be any setting. If the signs of setting are observed, reject the paste and start with a fresh specimen.



Prepare the trial paste with varying percentages of water until plunger penetrates to a point 5 to 7 mm from the bottom of the Vicat Mould, which is read on the scale. Express the water required as percentage by weight of the dry cement. The apparatus shall be free from vibrations during the test.

Determination Of Initial Setting Time

1. Specimen Preparation

Prepare a neat cement paste by gauging the cement with 0.85 times the water required to give a paste of a standard consistency in the manner and under the conditions already described above, preferably in an atmosphere of 90% relative humidity. Remove the grease on the initial needle and start a stop watch at the time that water is added to the cement.

2. Test Procedure

Attach a needle for the determination of initial setting time to the movable bearing rod of the Vicat Apparatus. Place the glass plate on the base of the Vicat Apparatus. Lower the movable rod and rest the needle on the plate. Raise the rod and pin it. Note the reading (A) on the scale. Fill the mould, positioned on the glass plate, with cement paste.

In filling the mould, the operator's hands and the gauging trowel shall alone be used. Place the mould with the test block on the glass plate on the base of the Vicat apparatus under the bearing rod to which the initial needle is attached. Slowly bring down the needle to the surface of the cement paste and quickly release. (When AIM-394 Vicat Apparatus with dash pot is used, place the mould filled with cement paste and the glass plate on the base of the Vicat Needle Apparatus).

Raise the plunger of the dashpot, bring it in contact with the top cap of the movable rod, remove the pin holding the movable bearing rod and bring down the initial needle attached to the movable bearing rod to the surface of the cement paste and quickly release it by pushing the plunger of the dashpot down by hand).

Make a note of the reading on the scale. Similarly, take the penetration reading at a regular interval, say at every 10 minutes. Be vigilant as the setting time approaches and observe the penetration at smaller intervals. Clean the needle from the sticking cement paste soon after it is withdrawn. Note the time when the needle rests, about a point 5 ± 0.5 mm from the original reading (A) which corresponds to the bottom of the Vicat Mould. This gives the initial setting time.

Determination Of Final Setting Time

Test Procedure

Remove the initial needle and fit in the final needle with the hollowed out metallic attachment to the bearing rod. Slowly lower the needle into contact with the surface of the paste and apply the needle gently to the surface (while AIM-394 Vicat Needle Apparatus with dashpot employ the plunger of the dashpot and set up the needle as described under the 'Determination of Initial Setting Time Test Procedure'). Repeat the same operation at regular intervals of time until the needle makes a mark on the surface of the test block while the metallic attachment fails to do so. Make a note of the time.

The time elapsed between the time when water is added to the dry material and the time at which the needle makes an impression on the surface while the attachment fails to do so, when upon applying the needle gently to the surface, gives the final setting time.

In the event of scum forming on the surface of the test block, the under-side of the test block may be used for determining the final test.

Take at least three specimens, the mean of these specimens is the final setting time.

Specifications for Setting Times of Various Types of Cements

S.No.	Type of Cement	Initial Setting time not less than	Final Setting time not more than
1.	Ordinary Portland	30 minutes	600 minutes
2.	Rapid Hardening Portland	30 minutes	600 minutes
3.	Low Heat Portland	60 minutes	600 minutes

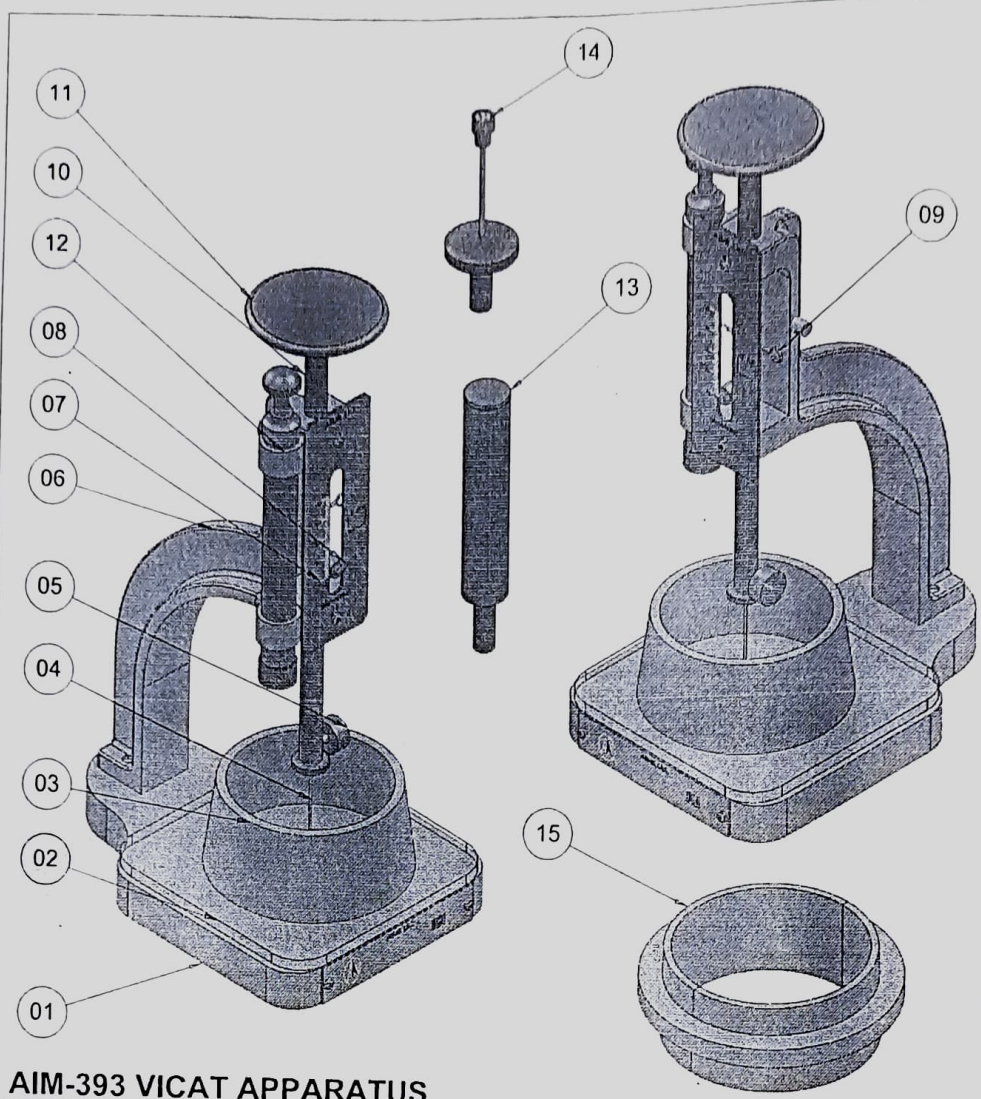


Precautions

1. The apparatus should remain free from vibrations during the test.
2. Care should be taken that the initial needle is not bent and is fixed vertically straight into the movable rod.
3. The needles and the plunger should be kept free from the adhering cement, as the cement adhering on the sides may retard penetration and cement at the bottom may increase.
4. All the tests should be carried out at $27^{\circ}\text{C} \pm 2^{\circ}\text{C}$ and 90% relative humidity.
5. Petroleum Jelly should be applied to the mould.

Maintenance

1. The bearing rod should be oiled periodically.
2. All needles should be kept in their respective cases after they are cleaned and greased.



AIM-393 VICAT APPARATUS

**AIM-394 VICAT APPARATUS
WITH DASHPOT**

S.NO.	DESCRIPTION
01	BASE
02	GLASS PLATE
03	VICAT MOULD
04	INITIAL NEEDLE
05	CLAMP SCREW
06	FRAME
07	GRADUATED SCALE
08	INDICATOR

S.NO.	DESCRIPTION
09	RELEASE PIN
10	MOVEABLE ROD
11	CAP
12	DASH POT
13	PLUNGER
14	FINAL NEEDLE
15	SPLIT VICAT MOULD